

RMDL Assessment 2: Data Analysis Report

B287980

0.1 Background

In bilinguals, research often reflects that the mother tongue is more strongly associated with emotion, while the second language can be used to create emotional distance (Dewaele 2010). However, results also reflect that early simultaneous bilinguals demonstrate similar patterns of response to emotion words in both languages (HARRIS and Berko Gleason 2006). It appears that processing of emotion words in first and second languages varies with proficiency.

The Stroop effect is a cognitive interference effect from the Stroop task (Stroop 1935), whereby identification of a written colour word's ink colour is slower when the written word is printed in a conflicting colour: for example, the word 'blue' printed in green ink is more slowly identified than when it is printed in blue ink, due to interference from the automatic process of reading during colour identification. Subsequent research has led to several variations of the Stroop task. One such variant is the emotional Stroop task, in which participants identify the ink colour of emotionally salient and neutral words rather than colour words. The resulting interference effect is theorised to be the result of interference caused by the emotion stimuli, the process for which is automatic and immediate (Bargh 1997). In addition, the Stroop paradigm has been widely applied in bilingual contexts to examine patterns of interference across languages (Mägiste 1984). Research largely suggests that interference effects vary with language proficiency, with lower proficiency typically leading to a greater interference effect (Sutton et al. 2007). This may reflect reduced automaticity in lexical processing, leading to increased competition between linguistic and task-related processes. The following study builds on the theory of the emotion Stroop effect and bilingual Stroop effect, to investigate the effect of age of acquisition of emotional interference in English-French bilingual speakers.

0.1.1 Research Questions/Hypotheses

Research question: Does age of acquisition influence reaction time to emotion and to neutral words in a bilingual Stroop paradigm?

Research hypothesis: Hypothesis is based on the results of Sutton et al. (2007). In early bilinguals, a slower reaction time is predicted for emotion words than for neutral words in

both French and English conditions. For late bilinguals, a slower reaction time is predicted for emotion words than for neutral words only in the English condition. An interaction between word condition and language is therefore only expected in late bilinguals. Therefore, we expect that an emotion interference effect will negatively correlate with language proficiency, with early bilinguals showing faster reaction times than late bilinguals.

0.2 Methods

0.2.1 Participants

The study uses two participants groups: early simultaneous bilingual French-English speakers (attainment of both languages before the age of 11) and sequential French-English speakers (native English speakers who learnt French as a second language after the age of 11). The late bilingual group had 6 participants, and the early bilingual groups had 5 participants, so a total of 11 participants took part in the study.

In the late bilingual group, acquisition of French was between the ages of 11 - 15 (mean = 11), with fluency (mean = 71% range = 50 - 90%) achieved from the ages of 16 - 20 (mean = 18), with one participant describing themselves as not fluent in French. All participants had proficiency and/or fluency additionally in Italian, Spanish, German, and BSL or a combination of these languages. 1 participant had acquired Italian and English from birth, and had acquired L4 French after L3 German, while the remaining 5 participants had acquired French as an L2. On average, participants were exposed to English 79.5% of the time, French 8.4% of the time, and other languages 12.1% of the time. 4 participants reported that they would choose to read in English over another language, while 1 participant reported a 50% preference for French and English, and 1 participant reported that they would choose to read in Italian.

In the early bilingual group, 4 participants were simultaneous learners of French and English, meaning they acquired them simultaneously from birth, while 1 participant acquired French from birth and English from age 4, and 1 participant was a native French speaker who acquired fluency in English from age 10. All participants had proficiency and/or fluency additionally in Spanish, German and Russian, or a combination of these languages. On average, participants were exposed to English 73.3% and to French 23% of the time, and to another language 3.7% of the time. 4 participants reported that they would choose to read in English, while 1 reported that they would choose to read in French.

All participants were aged between 19 - 28 and are university educated, and currently reside in the United Kingdom or the United States.

0.2.1.1 Table 1 - Participant characteristics

Participant group	Age of acquisition of French (mean, range)	Self-rated fluency level in French (mean, range)	Age at which fluency in French achieved (mean, range)	Current language exposure
Late bilinguals	11, 11 - 15	71%, 50 - 90%	18, 16 - 20	English = 79.5% French = 8.4% Other = 12.1%
Early bilinguals	1, 0-4	N/A	N/A	English = 73.3% French = 23% Other = 3.7%

0.2.2 Materials

Language history questionnaire

Firstly, all participants completed a language history questionnaire (Marian, Blumenfeld, and Kaushanskaya n.d.), which collects information on their language acquisition background and self-proficiency ratings. The information is presented below in Figure 1 (above).

Stroop task

The Stroop task assessment was built using the materials adapted from Sutton et al. (2007). In the original study, the word list was composed of two different types of words: negative emotion words (e.g. anger) and neutral (non-emotion related) words which belonged to the category ‘nautical nouns’ (e.g. boat). The English words were repeated in Spanish. Sutton et al. (2007) highlight that the chosen words were high in arousal and low in valence, and matched for frequency and length. Therefore, the current study uses the same English word list but with French translated equivalents (rather than Spanish) in order to build on the results of Sutton et al. (2007) and investigate a different language combination. The complete word list (French & English, emotion and neutral items) was repeated in blue ink and in green ink. The word list was therefore divided into three blocks: language (French/English), word type (neutral/emotion) and ink colour (blue/green). Due to the small scale of the study, the translated French words were not matched for frequency and length.

Participants saw a total of 32 items, as items were counterbalanced across colours and languages: half of the emotion items appeared in English and half in French, and the same applied to neutral items. The items were not repeated across languages (for example, if a participant saw the English word ‘boat’ they would not see the French equivalent ‘bateau’). Items were

also counterbalanced across colour: each word was presented once, in one colour, and not repeated in the second colour. In summary, participants saw eight neutral English items (4 in green, 4 in blue), eight emotion English items (4 in green, 4 in blue), eight neutral French items (4 in green, 4 in blue), and eight emotion French items (4 in green, 4 in blue). Items were blocked by language, and the order that the blocks were presented in was pseudorandomised and counterbalanced across participants.

A complete list of stimuli can be found in the Appendix.

0.2.3 Procedure

The Stroop task was conducted online via Testable, with each participant completing it on their own device. Participants first completed a practice session, consisting of a total of 6 neutral items, which were in a distinct semantic category to the critical trial neutral items. In the practice condition, the word appeared on the screen for 3 seconds. Participants were instructed to press the ‘L’ key if the word appeared in green ink, and the ‘A’ key if the word appeared in blue ink. Instructions only appeared in English. For the practice trials, the key/colour reminder prompt appeared on the screen alongside the practice word. Participants were not given feedback on the practice condition if their answer was correct. Following the practice condition, the participants immediately completed the critical trial conditions. They were instructed that in the critical condition, the word would appear on screen for less time - a total of 1.7 seconds - and the prompt would not appear on the screen alongside the critical word.

0.2.4 Statistical Analysis

In order to answer the research question, *does age of acquisition influence reaction time to emotion and to neutral words in a bilingual Stroop paradigm*, age of acquisition will be treated as a categorical variable of early bilinguals and late bilinguals. The bilingual Stroop paradigm is operationalised as language of stimuli (English, French).

Reaction times in milliseconds for each trial will be analysed using mixed-effects regression modelling. Summary measures for mean and standard deviation of the raw data will be reported. The study will then employ a 3x3 mixed factorial design of lognormal regression, with group (early vs late) as between-participant fixed effects and condition (emotion vs neutral) and language of stimuli (French vs English) as within-participant random effects. The posterior means, standard deviations and credible intervals will then be reported.

0.2.5 Data Availability

Anonymised participant data (Stroop task performance) is available on the following GitHub link: <https://github.com/KatiePedrotti/rmdl-template>

To respect participant privacy, language questionnaires are not available.

0.3 Data analysis

0.3.1 Reading and processing the data

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ggplot2)
library(brms)
```

```
Loading required package: Rcpp
Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').
```

```
Attaching package: 'brms'
```

```
The following object is masked from 'package:stats':
```

```
ar
```

```
library(dplyr)
library(posterior)
```

```
This is posterior version 1.6.1
```

Attaching package: 'posterior'

The following objects are masked from 'package:stats':

mad, sd, var

The following objects are masked from 'package:base':

%in%, match

```
library(tidybayes)
```

Warning: package 'tidybayes' was built under R version 4.5.3

Attaching package: 'tidybayes'

The following objects are masked from 'package:brms':

dstudent_t, pstudent_t, qstudent_t, rstudent_t

```
alldata = list.files(  
  path = "./data/",  
  pattern = "*.csv$",  
  recursive = TRUE ,  
  full.names = TRUE  
)  
  
compiledlist <- lapply(alldata, function(x) {  
  read_csv(x, skip = 3)  
})
```

Rows: 40 Columns: 26

-- Column specification -----

Delimiter: ","

chr (12): type, condition1, stim1, textColor, stimFormat, subjectGroup, keyb...

dbl (14): id, rowNo, ITI, presTime, random, ITI_ms, ITI_f, ITI_fDuration, pr...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```

Rows: 40 Columns: 26
-- Column specification -----
Delimiter: ","
chr (12): type, condition1, stim1, textColor, stimFormat, subjectGroup, keyb...
dbl (14): id, rowNo, ITI, presTime, random, ITI_ms, ITI_f, ITI_fDuration, pr...

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i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 40 Columns: 26
-- Column specification -----
Delimiter: ","

```

```
chr (12): type, condition1, stim1, textColor, stimFormat, subjectGroup, keyb...
dbl (14): id, rowNo, ITI, presTime, random, ITI_ms, ITI_f, ITI_fDuration, pr...

i Use `spec()` to retrieve the full column specification for this data.
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Rows: 40 Columns: 26
-- Column specification -----
Delimiter: ","
chr (12): type, condition1, stim1, textColor, stimFormat, subjectGroup, keyb...
dbl (14): id, rowNo, ITI, presTime, random, ITI_ms, ITI_f, ITI_fDuration, pr...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 40 Columns: 26
-- Column specification -----
Delimiter: ","
chr (12): type, condition1, stim1, textColor, stimFormat, subjectGroup, keyb...
dbl (14): id, rowNo, ITI, presTime, random, ITI_ms, ITI_f, ITI_fDuration, pr...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 40 Columns: 26
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Delimiter: ","
chr (12): type, condition1, stim1, textColor, stimFormat, subjectGroup, keyb...
dbl (14): id, rowNo, ITI, presTime, random, ITI_ms, ITI_f, ITI_fDuration, pr...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 40 Columns: 26
-- Column specification -----
Delimiter: ","
chr (12): type, condition1, stim1, textColor, stimFormat, subjectGroup, keyb...
dbl (14): id, rowNo, ITI, presTime, random, ITI_ms, ITI_f, ITI_fDuration, pr...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
finaldata <- bind_rows(compiledlist) |>
  filter(condition1 != "practise") |>
  mutate(
    early_late = case_when(
```



```

    substring(id,1,1) == "1"~"early",
    substring(id,1,1) == "2"~"late",
    TRUE ~""
  )
)

```

As the data were in separate files and lacking identifying columns, I firstly used `list.files()` to list all the available results files, and `lapply()` to compile them together, with `read_csv(x, skip=3)` in order to skip the first 3 rows of each dataset which did not contain any data. Then I used `bind_rows` to read in this data, and filtered out the 6 practise conditions using `filter()`. I then used `mutate()` in order to create a column which classified the participants as early or late bilinguals based on their participant number (1 were early bilinguals, 2 were late bilinguals).

```

newfinaldata <- finaldata |>
  separate_wider_delim(
    condition1,
    delim = "_",
    names = c("Condition", "Language")
  )

```

```

newfinaldata$Language <- recode(newfinaldata$Language,
                                "eng" = "English",
                                "fr"  = "French")

newfinaldata$early_late <- recode(newfinaldata$early_late,
                                "early" = "Early bilingual",
                                "late"  = "Late bilingual")

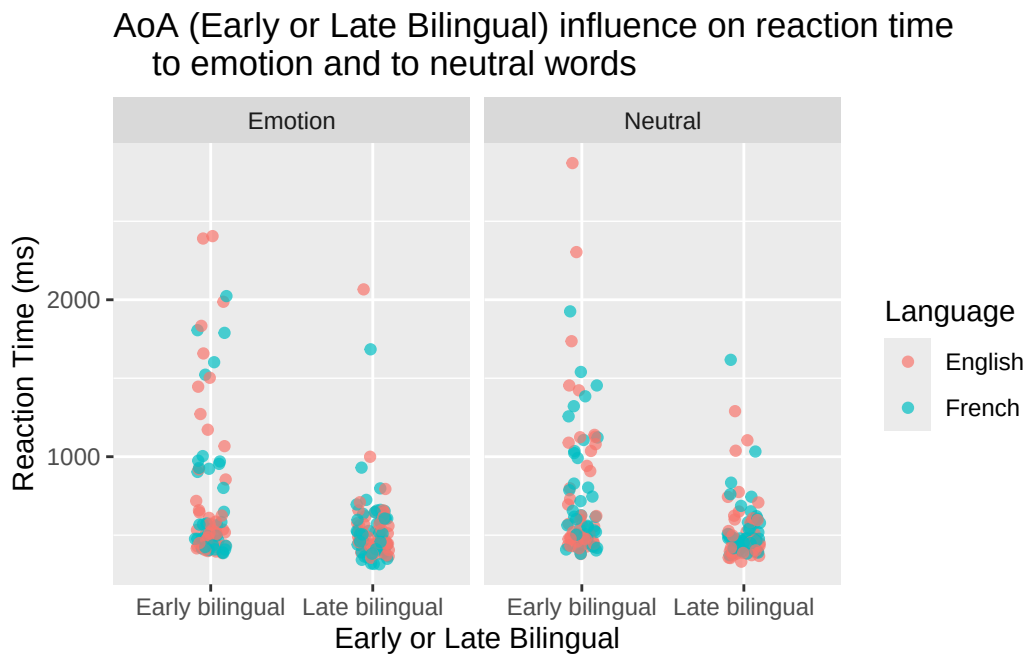
newfinaldata$Condition <- recode(newfinaldata$Condition,
                                "emotion" = "Emotion",
                                "neutral"  = "Neutral")

```

I used `separate_wider_delim()` in order to create new columns which identified the condition (emotion and neutral) and language (French and English), and re-coded the column names accordingly. The relevant columns are therefore Condition (emotion/neutral stimuli), Language (French/English stimuli), and Early_Late (early or late bilingual), as well as RT (reaction time).

0.3.2 Plotting the data

```
newfinaldata |>
  ggplot(aes(early_late, RT, colour = Language)) +
  geom_jitter(alpha = 0.7, width = 0.1) +
  facet_wrap(~ Condition) +
  labs(
    x = "Early or Late Bilingual",
    y = "Reaction Time (ms)",
    title = "AoA (Early or Late Bilingual) influence on reaction time
to emotion and to neutral words")
```



0.3.2.1 Figure 1 - Age of acquisition (early or late bilingual) effect on reaction time (RT) across condition (emotion and neutral stimuli) and language (French and English).

I used a jitter plot to represent the raw data, since it combines numeric (reaction times) and categorical variables. The jitter plot is faceted by Condition (neutral and emotion stimuli), with the x-axis showing Participant group (early or late bilingual) and the y-axis showing Reaction Time (ms). The colours differentiate the Language of the stimuli (French = blue, English = red).

The plot shows firstly in the emotion condition, that late bilinguals showed less variation and therefore are more densely represented than the early bilinguals. There is minimal variation across both participant groups between languages (English compared to French).

In the neutral condition, there is slightly more variation across languages, with late bilinguals appearing to perform very slightly faster in English than in French, and early bilinguals' reaction times showing more variation compared to late bilinguals in this condition. The plot reflects a slightly faster reaction time across conditions and across language for late bilinguals.

There are some outliers across condition, group and language, but these appear to be more pronounced in the early bilinguals compared to the late bilinguals.

The raw data is further described in the summary measures (below).

0.3.3 Summary of the data

```
newfinaldata |>
  group_by(Condition, Language, early_late) |>
  summarise(
    mean = mean(RT), sd = sd(RT))
```

`summarise()` has grouped output by 'Condition', 'Language'. You can override using the `.groups` argument.

```
# A tibble: 8 x 5
# Groups:   Condition, Language [4]
  Condition Language early_late    mean    sd
  <chr>      <chr>    <chr>      <dbl> <dbl>
1 Emotion   English   Early bilingual  818.  561.
2 Emotion   English   Late bilingual  542.  255.
3 Emotion   French    Early bilingual  726.  440.
4 Emotion   French    Late bilingual  533.  216.
5 Neutral   English   Early bilingual  806.  537.
6 Neutral   English   Late bilingual  516.  196.
7 Neutral   French    Early bilingual  756.  377.
8 Neutral   French    Late bilingual  537.  204.
```

In order to investigate if age of acquisition affects reaction time to emotion and neutral words in the bilingual Stroop task, I analysed participant reaction times across condition (neutral vs. emotion) and language (French vs. English).

Across conditions, late bilinguals showed faster mean reaction times than early bilinguals.

In the emotion condition, early bilinguals had a faster reaction time in French (mean = 726.38ms, SD = 440.26) than in English (mean = 817.83ms, SD = 561.09), while the late bilinguals showed the reverse pattern, demonstrating slightly faster reaction times in French (mean = 533.33ms, SD = 215.51) than in English (mean = 542.31ms, SD = 255.28). Overall in the emotion condition, the late bilinguals had faster reaction times in both languages than the early bilinguals.

A similar pattern can be observed in the neutral condition. In language conditions, early bilinguals had a faster reaction time in French (mean = 755.85ms, SD = 376.74ms) than in English (mean = 805.58ms, SD = 536.91ms) in the neutral condition, while late bilinguals had a faster reaction time in English (mean = 515.77ms, SD = 196.49ms) than in French (mean = 536.77ms, SD = 204.00ms) in the neutral condition. Therefore overall in the neutral condition, late bilinguals showed faster reaction times in both languages compared to early bilinguals.

Comparing conditions, early bilinguals showed a slightly faster reaction time in the emotion condition in English (English: mean = 817.83ms, SD = 561.08ms) compared to the neutral condition in English (English: mean = 805.58ms, SD = 536.91ms) but showed the reverse pattern in the emotion condition in French (emotion: mean = 726.38ms, SD = 440.26, neutral: mean = 755.85ms, SD = 376.74ms).

The late bilinguals showed the reverse pattern, demonstrating a slower reaction time in the emotion condition in English (mean = 542.31, SD = 255.28) compared to the neutral condition in English (mean = 515.77ms, SD = 196.49ms), but the same pattern as the early bilinguals across the French conditions, such that they performed slightly faster in the French emotion condition (mean = 533.33ms, SD = 215.51) than the French neutral condition (mean = 536.77ms, SD = 204.00ms), but to a lesser extent than the early bilinguals demonstrated.

Taken together, late bilinguals had faster reaction times than early bilinguals, with early bilinguals showing a faster reaction time to emotion words in French (compared to neutral condition), and late bilinguals showing a faster reaction time in the neutral condition in English. The differing patterns across participant groups in the raw data therefore suggest that age of acquisition may affect reaction time to emotion and neutral words across condition and language, such that late bilinguals showed a faster reaction time across both language and condition.

0.3.4 Regression modelling

I used Bayesian log-normal regression model for the outcome variable reaction time (RT) using brms. This distribution allows positive-only values, continuous variables, and is able to account for right-skewed data. This was fitted with Group (early vs late bilinguals) as between-participant fixed effects, and Condition (emotion vs neutral) and Language (French vs English) as within-participant random effects, and including Participant ID as random effect

to account for inter-participant variability. I then obtained summary measures. As the values are reported as logged reaction time, I used `exp()` on the relevant values to obtain the unlogged reaction times.

```
brm_newfinaldata <- brm(
  RT ~ early_late * Language * Condition +
    (Language + Condition | id ),
  family = lognormal,
  data = newfinaldata,
  seed = 456,
  cores = 4,
  file = "cache/brmfinaldata",
  control = list(adapt_delta = 0.99, max_treedepth = 15)
)
```

```
summary(brm_newfinaldata)
```

```
Family: lognormal
Links: mu = identity
Formula: RT ~ early_late * Language * Condition + (Language + Condition | id)
Data: newfinaldata (Number of observations: 352)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Multilevel Hyperparameters:

~id (Number of levels: 11)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat
sd(Intercept)	0.40	0.12	0.24	0.69	1.00
sd(LanguageFrench)	0.05	0.04	0.00	0.15	1.00
sd(ConditionNeutral)	0.06	0.05	0.00	0.18	1.00
cor(Intercept,LanguageFrench)	-0.32	0.46	-0.94	0.70	1.00
cor(Intercept,ConditionNeutral)	-0.12	0.45	-0.86	0.79	1.00
cor(LanguageFrench,ConditionNeutral)	-0.07	0.49	-0.89	0.85	1.00

	Bulk_ESS	Tail_ESS
sd(Intercept)	1410	1785
sd(LanguageFrench)	1939	1869
sd(ConditionNeutral)	1508	1335
cor(Intercept,LanguageFrench)	3895	2755
cor(Intercept,ConditionNeutral)	4083	2955
cor(LanguageFrench,ConditionNeutral)	2487	2824

Regression Coefficients:

	Estimate	Est.Error		
Intercept	6.54	0.18		
early_lateLatebilingual	-0.29	0.25		
LanguageFrench	-0.08	0.07		
ConditionNeutral	0.01	0.07		
early_lateLatebilingual:LanguageFrench	0.07	0.09		
early_lateLatebilingual:ConditionNeutral	-0.05	0.09		
LanguageFrench:ConditionNeutral	0.07	0.08		
early_lateLatebilingual:LanguageFrench:ConditionNeutral	-0.01	0.12		
	1-95% CI	u-95% CI	Rhat	
Intercept	6.17	6.92	1.00	
early_lateLatebilingual	-0.79	0.18	1.01	
LanguageFrench	-0.21	0.05	1.00	
ConditionNeutral	-0.12	0.14	1.00	
early_lateLatebilingual:LanguageFrench	-0.11	0.25	1.00	
early_lateLatebilingual:ConditionNeutral	-0.23	0.13	1.00	
LanguageFrench:ConditionNeutral	-0.10	0.24	1.00	
early_lateLatebilingual:LanguageFrench:ConditionNeutral	-0.24	0.23	1.00	
	Bulk_ESS	Tail_ESS		
Intercept	965	1898		
early_lateLatebilingual	981	1825		
LanguageFrench	2101	2546		
ConditionNeutral	2164	2434		
early_lateLatebilingual:LanguageFrench	2167	2787		
early_lateLatebilingual:ConditionNeutral	2229	2620		
LanguageFrench:ConditionNeutral	2156	2769		
early_lateLatebilingual:LanguageFrench:ConditionNeutral	2232	2687		

Further Distributional Parameters:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.28	0.01	0.26	0.30	1.00	5596	3054

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
exp(6.17) #Intercept_95%
```

```
[1] 478.1861
```

```
exp(6.92) #Intercept_95%
```

```
[1] 1012.32
```

```
summary(brm_newfinaldata, prob = 0.8)
```

```
Family: lognormal
Links: mu = identity
Formula: RT ~ early_late * Language * Condition + (Language + Condition | id)
Data: newfinaldata (Number of observations: 352)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Multilevel Hyperparameters:

~id (Number of levels: 11)

	Estimate	Est.Error	1-80% CI	u-80% CI	Rhat
sd(Intercept)	0.40	0.12	0.28	0.55	1.00
sd(LanguageFrench)	0.05	0.04	0.01	0.11	1.00
sd(ConditionNeutral)	0.06	0.05	0.01	0.13	1.00
cor(Intercept,LanguageFrench)	-0.32	0.46	-0.85	0.37	1.00
cor(Intercept,ConditionNeutral)	-0.12	0.45	-0.71	0.53	1.00
cor(LanguageFrench,ConditionNeutral)	-0.07	0.49	-0.71	0.63	1.00

	Bulk_ESS	Tail_ESS
sd(Intercept)	1410	1785
sd(LanguageFrench)	1939	1869
sd(ConditionNeutral)	1508	1335
cor(Intercept,LanguageFrench)	3895	2755
cor(Intercept,ConditionNeutral)	4083	2955
cor(LanguageFrench,ConditionNeutral)	2487	2824

Regression Coefficients:

	Estimate	Est.Error
Intercept	6.54	0.18
early_lateLatebilingual	-0.29	0.25
LanguageFrench	-0.08	0.07
ConditionNeutral	0.01	0.07
early_lateLatebilingual:LanguageFrench	0.07	0.09
early_lateLatebilingual:ConditionNeutral	-0.05	0.09
LanguageFrench:ConditionNeutral	0.07	0.08
early_lateLatebilingual:LanguageFrench:ConditionNeutral	-0.01	0.12

	1-80% CI	u-80% CI	Rhat
--	----------	----------	------

Intercept	6.31	6.76	1.00
early_lateLatebilingual	-0.59	0.01	1.01
LanguageFrench	-0.16	0.00	1.00
ConditionNeutral	-0.08	0.09	1.00
early_lateLatebilingual:LanguageFrench	-0.05	0.18	1.00
early_lateLatebilingual:ConditionNeutral	-0.17	0.07	1.00
LanguageFrench:ConditionNeutral	-0.04	0.17	1.00
early_lateLatebilingual:LanguageFrench:ConditionNeutral	-0.15	0.14	1.00

	Bulk_ESS	Tail_ESS
Intercept	965	1898
early_lateLatebilingual	981	1825
LanguageFrench	2101	2546
ConditionNeutral	2164	2434
early_lateLatebilingual:LanguageFrench	2167	2787
early_lateLatebilingual:ConditionNeutral	2229	2620
LanguageFrench:ConditionNeutral	2156	2769
early_lateLatebilingual:LanguageFrench:ConditionNeutral	2232	2687

Further Distributional Parameters:

	Estimate	Est.Error	l-80% CI	u-80% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.28	0.01	0.26	0.29	1.00	5596	3054

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
exp(6.31) #Intercept_80%
```

```
[1] 550.0449
```

```
exp(6.76) #Intercept_80%
```

```
[1] 862.6422
```

```
summary(brm_newfinaldata, prob = 0.6)
```

```
Family: lognormal
Links: mu = identity
Formula: RT ~ early_late * Language * Condition + (Language + Condition | id)
Data: newfinaldata (Number of observations: 352)
```


Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
total post-warmup draws = 4000

Multilevel Hyperparameters:

~id (Number of levels: 11)

	Estimate	Est.Error	1-60% CI	u-60% CI	Rhat
sd(Intercept)	0.40	0.12	0.31	0.49	1.00
sd(LanguageFrench)	0.05	0.04	0.02	0.09	1.00
sd(ConditionNeutral)	0.06	0.05	0.02	0.10	1.00
cor(Intercept,LanguageFrench)	-0.32	0.46	-0.74	0.09	1.00
cor(Intercept,ConditionNeutral)	-0.12	0.45	-0.55	0.30	1.00
cor(LanguageFrench,ConditionNeutral)	-0.07	0.49	-0.54	0.40	1.00
	Bulk_ESS	Tail_ESS			
sd(Intercept)	1410	1785			
sd(LanguageFrench)	1939	1869			
sd(ConditionNeutral)	1508	1335			
cor(Intercept,LanguageFrench)	3895	2755			
cor(Intercept,ConditionNeutral)	4083	2955			
cor(LanguageFrench,ConditionNeutral)	2487	2824			

Regression Coefficients:

	Estimate	Est.Error			
Intercept	6.54	0.18			
early_lateLatebilingual	-0.29	0.25			
LanguageFrench	-0.08	0.07			
ConditionNeutral	0.01	0.07			
early_lateLatebilingual:LanguageFrench	0.07	0.09			
early_lateLatebilingual:ConditionNeutral	-0.05	0.09			
LanguageFrench:ConditionNeutral	0.07	0.08			
early_lateLatebilingual:LanguageFrench:ConditionNeutral	-0.01	0.12			
	1-60% CI	u-60% CI	Rhat		
Intercept	6.39	6.67	1.00		
early_lateLatebilingual	-0.49	-0.09	1.01		
LanguageFrench	-0.14	-0.03	1.00		
ConditionNeutral	-0.05	0.06	1.00		
early_lateLatebilingual:LanguageFrench	-0.01	0.14	1.00		
early_lateLatebilingual:ConditionNeutral	-0.13	0.03	1.00		
LanguageFrench:ConditionNeutral	-0.00	0.13	1.00		
early_lateLatebilingual:LanguageFrench:ConditionNeutral	-0.10	0.09	1.00		
	Bulk_ESS	Tail_ESS			
Intercept	965	1898			
early_lateLatebilingual	981	1825			
LanguageFrench	2101	2546			

ConditionNeutral	2164	2434
early_lateLatebilingual:LanguageFrench	2167	2787
early_lateLatebilingual:ConditionNeutral	2229	2620
LanguageFrench:ConditionNeutral	2156	2769
early_lateLatebilingual:LanguageFrench:ConditionNeutral	2232	2687

Further Distributional Parameters:

	Estimate	Est.Error	l-60% CI	u-60% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.28	0.01	0.27	0.29	1.00	5596	3054

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
exp(6.39) #Intercept_60%
```

```
[1] 595.8566
```

```
exp(6.67) #Intercept_60%
```

```
[1] 788.3956
```

```
exp(-0.49) #early_lateLatebilingual_60%
```

```
[1] 0.6126264
```

```
exp(-0.09) #early_lateLatebilingual_60%
```

```
[1] 0.9139312
```

```
exp(-0.14) #LanguageFrench
```

```
[1] 0.8693582
```

```
exp(-0.03) #LanguageFrench
```

```
[1] 0.9704455
```

The model did converge, however the summary measures reflect a high degree of uncertainty.

At 95% credible interval, the model estimates that the mean reaction time for early bilinguals in the emotion and English condition is between 6.17 (lower bound) and 6.92 (upper bound) logged milliseconds, and 478.19ms (lower bound) and 1,0312.32ms (upper bound) unlogged milliseconds. All predictors cross zero at 95% credible interval, and therefore there is a high degree of uncertainty as to their effect size.

At 80% credible interval, all predictors still cross zero, but to a lesser extent. The mean reaction time predicted at the Intercept (english_emotion_early) is between 6.31 logged milliseconds and 6.76 logged milliseconds, and 550.04 and 862.64 unlogged milliseconds.

Finally, at 60% credibility, the predicted reaction time at the intercept is between 6.17 and 6.92 logged milliseconds, or between 595.86 and 788.38 unlogged milliseconds at the intercept (english_emotion_early). At 60% credibility, two predictors now show a decrease in log milliseconds. Firstly, english_emotion_late show an effect size between -0.49 and -0.09 (logged) or 0.61 to 0.91 (unlogged), suggesting that late bilinguals are predicted to be faster than early bilinguals in the emotion/English condition. Secondly, french_emotion_early show an effect size between -0.14 and -0.03 (logged) or 0.86 to 0.97 (unlogged), suggesting that early bilinguals in the emotion condition are likely to show a faster reaction time in French than in English. However, the remaining predictors cross zero and there remains a high degree of uncertainty regarding how these predictors interact across the model.

After obtaining summary measures at 95%, 80% and 60% credible intervals, I obtained the model draws, implementing the relevant predictors and interactions based on the b_Intercept as english_emotion_early. I then had select() the relevant columns and pivot_longer() in order to make new columns for these predictors, which I could then plot into the graph (below).

```
draws_brm_newfinaldata <- as_draws_df(brm_newfinaldata)

frame_draws_brm_newfinaldata <- draws_brm_newfinaldata |>
  mutate(
    eng_emotion_early = b_Intercept,
    fr_emotion_early = b_Intercept + b_LanguageFrench,
    eng_emotion_late = b_Intercept + b_early_lateLatebilingual,
    eng_neutral_early = b_Intercept + b_ConditionNeutral,
    fr_emotion_late = b_Intercept + b_early_lateLatebilingual
    + b_LanguageFrench + `b_early_lateLatebilingual:LanguageFrench`,
    eng_neutral_late = b_Intercept + b_ConditionNeutral +
      b_early_lateLatebilingual + `b_early_lateLatebilingual:ConditionNeutral`,
    fr_neutral_early = b_Intercept + b_LanguageFrench + b_ConditionNeutral + `b_LanguageFrench:ConditionNeutral`,
    fr_neutral_late = b_Intercept + b_ConditionNeutral + b_LanguageFrench +
      b_early_lateLatebilingual + `b_early_lateLatebilingual:LanguageFrench` +
      `b_LanguageFrench:ConditionNeutral` +
```

```

`b_early_lateLatebilingual:ConditionNeutral` +
`b_early_lateLatebilingual:LanguageFrench:ConditionNeutral`
) |>
select(eng_emotion_early, fr_emotion_early, eng_emotion_late,
       eng_neutral_early, fr_emotion_late, eng_neutral_late,
       fr_neutral_early, fr_neutral_late)

```

Warning: Dropping 'draws_df' class as required metadata was removed.

```

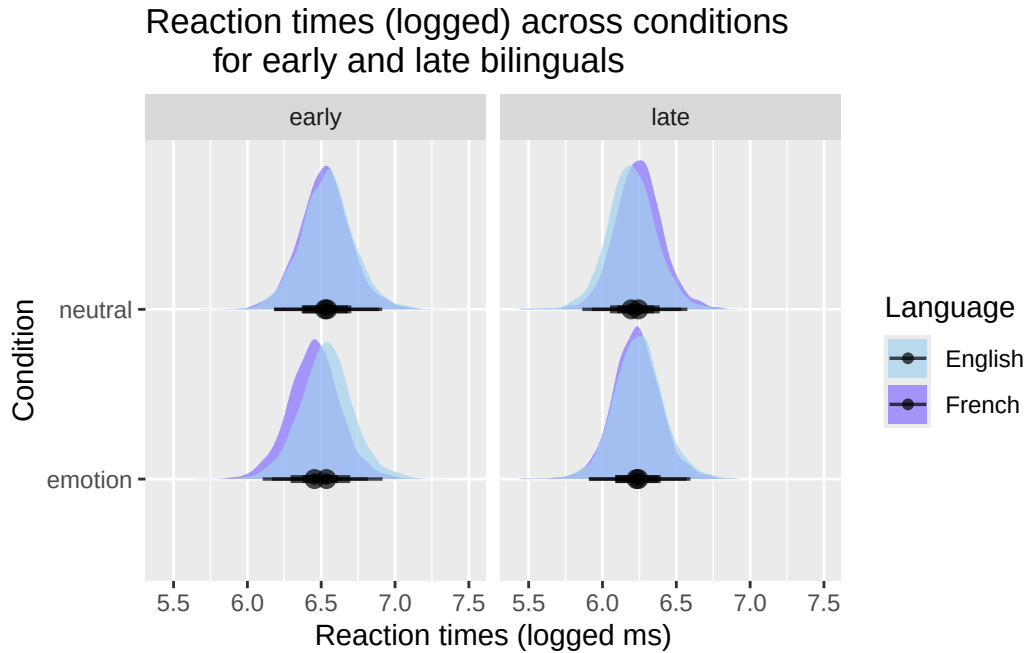
plot2 <- frame_draws_brm_newfinaldata |>
  select(eng_emotion_early, fr_emotion_early, eng_emotion_late,
         eng_neutral_early, fr_emotion_late, eng_neutral_late,
         fr_neutral_early, fr_neutral_late) |>
pivot_longer(
  eng_emotion_early:fr_neutral_late,
  names_to = "cond",
  values_to = "epred"
) |>
separate(cond, c("Language", "Condition", "Early_Late"))

```

```

plot2 |>
  ggplot(aes(epred, Condition, fill = Language )) +
  stat_halfeye(alpha = 0.7) +
  scale_fill_manual(
    labels = c("English", "French"),
    values = c("lightskyblue2", "slateblue1")
  ) + facet_grid(cols = vars(Early_Late)) +
  labs(title = "Reaction times (logged) across conditions
         for early and late bilinguals", x = "Reaction times (logged ms)")

```



0.3.4.1 Figure 2 - Age of acquisition (early or late bilingual) effect on logged reaction time (RT) across condition (emotion and neutral stimuli) and language (French and English).

I used a half-eye plot to demonstrate logged Reaction times on the x-axis, faceted by Participant group (early, late), and Condition on the y-axis (neutral, emotion), with colour differentiating language (English, French). The plot reflects the summary measures and model draws, showing very little variability between performance of early and late bilinguals across condition and across language. The most noticeable variation is in the emotion condition for early bilinguals, who appear to show a slightly faster reaction time in French than in English: the model did predict a negative effect on reaction time (i.e. faster reaction time) in these conditions at 60% credibility, however the summary measures showed this to still have a large degree of uncertainty.

0.4 Discussion

Write a brief discussion of the your model results in light of the research question.

0.5 Appendix

0.5.1 Stimuli

English - Emotion	French - Emotion	English - Neutral	French - Neutral
nervous	nerveux	engine	barre
grief	chagrin	boom	bôme
upset	contrarié	stern	poupe
greed	avidité	cabin	cabine
anxiety	anxiété	keel	quille
jealousy	jalousie	motor	moteur
lonely	seul	seat	siège
scared	effrayé	deck	pont
afraid	craintif	driver	conducteur
panic	panique	propoeller	hélice
fear	peur	floor	plancher
angry	énervé	anchor	ancre
disgusted	dégoûté	paddle	pagaie
depressed	déprimé	beam	barrot
mad	furieux	port	bâbord
troubled	tourmenté	rope	corde

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